MATH 345 Skeleton Notes
Unit 1: Vectors, matrices, and linear maps
Section 1.3: Matrix-vector product and linear maps

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Matrix-vector product

A vector can be written as a row or as a column. When a matrix acts on a vector in this course, we usually write the vector as a column:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

From now on, when we write $\mathbf{x} = (x_1, \dots, x_n)$, $\mathbf{x}$ is interpreted as a column vector.
If A is an $m \times n$ matrix, then A has n columns. We write

$$A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}$$

where each $\mathbf{a}_j$ is a column vector in $\mathbb{R}^m$.

Definition: Matrix-vector product
Let $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}$ be an $m \times n$ matrix and let $\mathbf{x} = (x_1, \dots, x_n)$ be a column vector in $\mathbb{R}^n$. The matrix-vector product is

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

Activity: Computing a matrix-vector product (U1-LO4)

Compute $A\mathbf{x}$ where

$$A = \begin{bmatrix} -1 & 4 & -5 \\ 3 & 1 & -2 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}.$$

Using columns,

$$\begin{aligned} A\mathbf{x} &= 2\begin{bmatrix} -1 \\ 3 \end{bmatrix} - 3\begin{bmatrix} 4 \\ 1 \end{bmatrix} + 4\begin{bmatrix} -5 \\ -2 \end{bmatrix} \\ &= \begin{bmatrix} 2(-1) - 3(4) + 4(-5) \\ 2(3) - 3(1) + 4(-2) \end{bmatrix} \\ &= \begin{bmatrix} -34 \\ -5 \end{bmatrix}. \end{aligned}$$

Theorem: Row dot-product view

If A is an $m \times n$ matrix with rows $\text{row}_1(A), \ldots, \text{row}_m(A)$ as n-vectors, then

$$A\mathbf{x} = \begin{bmatrix} \text{row}_1(A) \cdot \mathbf{x} \\ \text{row}_2(A) \cdot \mathbf{x} \\ \vdots \\ \text{row}_m(A) \cdot \mathbf{x} \end{bmatrix}.$$

Why is this true?

Write $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$, where $\mathbf{a}_j = (a_{1j}, \ldots, a_{mj})$, and let $\mathbf{x} = (x_1, \ldots, x_n)$. Since

$$A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n,$$

its i-th entry is

$$x_1 a_{i1} + \cdots + x_n a_{in} = \text{row}_i(A) \cdot \mathbf{x}.$$

Stacking these entries gives the stated formula.

Activity: Computing the same product using rows (U1-LO4)

We revisit Computing a matrix-vector product (U1-LO4) using row dot products. Compute $A\mathbf{x}$ where

$$A = \begin{bmatrix} -1 & 4 & -5 \\ 3 & 1 & -2 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}.$$

Take the dot product of each row of A with $\mathbf{x}$:

$$\begin{aligned} A\mathbf{x} &= \begin{bmatrix} (-1, 4, -5) \cdot (2, -3, 4) \\ (3, 1, -2) \cdot (2, -3, 4) \end{bmatrix} \\ &= \begin{bmatrix} (-1)(2) + 4(-3) + (-5)(4) \\ 3(2) + 1(-3) + (-2)(4) \end{bmatrix} \\ &= \begin{bmatrix} -34 \\ -5 \end{bmatrix}. \end{aligned}$$

Each dot product gives one component of the same output vector obtained using columns.

Theorem: Properties of matrix-vector products

Let A and B be $m \times n$ matrices, let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, and let r be a scalar. Then

- $A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y}$,
- $A(r\mathbf{x}) = r(A\mathbf{x}) = (rA)\mathbf{x}$,
- $(A + B)\mathbf{x} = A\mathbf{x} + B\mathbf{x}$.

Why is this true?

We prove the first property. Write $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$, $\mathbf{x} = (x_1, \ldots, x_n)$, and $\mathbf{y} = (y_1, \ldots, y_n)$. By the definition above,

$$\begin{aligned} A(\mathbf{x} + \mathbf{y}) &= (x_1 + y_1)\mathbf{a}_1 + \cdots + (x_n + y_n)\mathbf{a}_n \\ &= A\mathbf{x} + A\mathbf{y}. \end{aligned}$$

Note: Shape habit for matrix-vector products

Before forming $A\mathbf{x}$, check that the number of columns of A equals the number of entries of $\mathbf{x}$. If A is $m \times n$ and $\mathbf{x} \in \mathbb{R}^n$, then $A\mathbf{x} \in \mathbb{R}^m$: its m entries come from the m row dot products above.

Note: Rows measure; columns contribute

A matrix-vector product has two complementary readings. In the row view, each row measures the input by taking a dot product with $\mathbf{x}$. In the column view,

$$A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n,$$

so the input coordinates tell how much each column contributes to the output.

Examples of matrix-vector products

The same product $A\mathbf{x}$ can represent different operations, depending on what the rows and columns of A mean. The next activities show three common patterns: scoring, selecting, and measuring changes.

Activity: Scoring objects by features (U1-LO4)

Let the rows of X represent three objects with two features:

$$X = \begin{bmatrix} 3 & 1 \\ 1 & 4 \\ 2 & 2 \end{bmatrix}, \qquad \mathbf{w} = (2, -1).$$

Compute $X\mathbf{w}$. Interpret the result.

We have

$$X\mathbf{w} = \begin{bmatrix} 3 & 1 \\ 1 & 4 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 2 \end{bmatrix}.$$

Each output is a score:

$$2(\text{first feature}) - (\text{second feature}).$$

Activity: A selector matrix (U1-LO4, U1-LO5)

Let

$$S = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \end{bmatrix}.$$

Compute $S\mathbf{x}$. What does S do?

We have

$$S\mathbf{x} = \begin{bmatrix} 1 \cdot 5 + 0 \cdot 6 + 0 \cdot 7 + 0 \cdot 8 \\ 0 \cdot 5 + 0 \cdot 6 + 1 \cdot 7 + 0 \cdot 8 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}.$$

The matrix S selects the first and third entries of $\mathbf{x}$.

Activity: A difference matrix (U1-LO4, U1-LO5)

Let

$$D = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 2 \\ 5 \\ 9 \end{bmatrix}.$$

Compute $D\mathbf{x}$. What does D measure?

We have

$$D\mathbf{x} = \begin{bmatrix} -1 \cdot 2 + 1 \cdot 5 + 0 \cdot 9 \\ 0 \cdot 2 + (-1) \cdot 5 + 1 \cdot 9 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

The entries are consecutive differences:

$$5 - 2 = 3, \qquad 9 - 5 = 4.$$

Thus D measures changes in a short time series.

Activity: Document ranking in matrix code (U1-LO4, U1-LO8)

In Word-count vectors (U1-LO1, U1-LO2), we ranked documents by cosine similarity one score at a time. The following code repeats that ranking using a data matrix whose rows are the document vectors.

```
import numpy as np

doc_names = np.array(["D1", "D2", "D3"])

X = np.array([
    [3., 0., 3.],
    [1., 1., 0.],
    [0., 2., 0.],
])

q = np.array([1., 0., 1.])
```

Activity: Document ranking in matrix code (U1-LO4, U1-LO8) (continued)

```
scores = (X @ q) / (
    np.linalg.norm(X, axis=1) * np.linalg.norm(q)
)
ranking = doc_names[np.argsort(scores)[::-1]]

scores, ranking
```

Output:

```
(array([1. , 0.5, 0. ]),
 array(['D1', 'D2', 'D3']))
```

1. What does the second row of `X` represent?
2. Which expression computes the three dot products $\mathbf{D}_i \cdot \mathbf{q}$?
3. What does `np.argsort(scores)[::-1]` do?

The rows of `X` are $\mathbf{D}_1$, $\mathbf{D}_2$, and $\mathbf{D}_3$. The product `X @ q` computes the three dot products with the query.

To understand `np.argsort(scores)[::-1]`, read it in two steps:

1. `np.argsort(scores)` returns the indices that put the scores in increasing order, rather

than the sorted score values. NumPy indices start at `0`. For scores `[1, 0.5, 0]`, it returns `[2, 1, 0]`: the smallest score is at index `2`, the next at index `1`, and the largest at index `0`.

2. `[::-1]` reverses that index array. A slice has the form `[start:stop:step]`; leaving the endpoints blank and using step `-1` traverses the whole array backward. Thus `[2, 1, 0]` becomes `[0, 1, 2]`, which lists the indices from largest score to smallest.

The assignment to `ranking` uses these indices to select entries from `doc_names`. We could write the same calculation in two lines:

```
order = np.argsort(scores)[::-1]
ranking = doc_names[order]
```

Here `order` is `[0, 1, 2]`, so `ranking` contains `doc_names[0]`, `doc_names[1]`, and `doc_names[2]`, in that order. Thus the document names are arranged from highest score to

lowest. If the scores changed, the indices could select the names in a different order.
The displayed output `array(['D1', 'D2', 'D3'])` describes a one-dimensional NumPy array containing three text strings.
The scores agree with Word-count vectors (U1-LO1, U1-LO2): $1, \frac{1}{2}, 0$.

Example: The attention activity in matrix form

In Gathering context for the bird position (U1-LO1, U1-LO8), we computed the scores

$$\mathbf{q} \cdot \mathbf{k}_{\text{small}}, \qquad \mathbf{q} \cdot \mathbf{k}_{\text{red}}, \qquad \mathbf{q} \cdot \mathbf{k}_{\text{bird}}$$

one at a time. Matrix-vector multiplication computes the same scores at once.
Put the key vectors as the rows of

$$K = \begin{bmatrix} \mathbf{k}_{\text{small}}^T \\ \mathbf{k}_{\text{red}}^T \\ \mathbf{k}_{\text{bird}}^T \end{bmatrix}.$$

Then

$$K\mathbf{q} = \begin{bmatrix} \mathbf{k}_{\text{small}} \cdot \mathbf{q} \\ \mathbf{k}_{\text{red}} \cdot \mathbf{q} \\ \mathbf{k}_{\text{bird}} \cdot \mathbf{q} \end{bmatrix}.$$

For the numbers in Gathering context for the bird position (U1-LO1, U1-LO8),

$$K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{q} = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

so

$$K\mathbf{q} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

Activity: The same attention calculation in code (U1-LO4, U1-LO8)

The following code performs the calculation from the token attention activity.

```
import numpy as np

tokens = np.array(["small", "red", "bird"])

q = np.array([1., 1.])

K = np.array([
    [1., 0.],
    [0., 1.],
    [1., 1.],
])

V = np.array([
    [4., 0.],
    [0., 4.],
    [4., 4.],
])

s = K @ q
alpha = s / s.sum()
out = alpha @ V

s, alpha, out
```

Activity: The same attention calculation in code (U1-LO4, U1-LO8) (continued)

Output:

```
(array([1., 1., 2.]),
 array([0.25, 0.25, 0.5 ]),
 array([3., 3.]))
```

1. Which line computes the token scores?
2. Which token receives the largest weight?
3. Which line forms the weighted average?
4. Why do the entries of `alpha` add to 1?

The line `s = K @ q` computes the three dot products. The token "bird" receives the largest weight. The line `out = alpha @ V` forms the weighted average of the rows of `V`. The entries of `alpha` add to 1 because the scores were divided by their sum.

Definition: Standard basis

The j-th standard basis vector e_j is the column vector with a 1 in position j and zeros elsewhere.

The term "basis" will be explained later in the course. For now, use the

concrete description of $\mathbf{e}_j$ above.

Fact

If $A = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_n \end{bmatrix}$, then $A\mathbf{e}_j = \mathbf{a}_j$. Applying a matrix to a standard basis vector picks out a column.

Why is this true?

The coordinates of $\mathbf{e}_j$ are all zero except for a 1 in position j. Thus the column formula for the matrix-vector product gives

$$A\mathbf{e}_j = \sum_{k=1}^{n} (\mathbf{e}_j)_k \mathbf{a}_k = \mathbf{a}_j.$$

Here $(\mathbf{e}_j)_k$ denotes the k-th coordinate of $\mathbf{e}_j$, which equals 1 when $k = j$ and 0 otherwise.

Example: The identity matrix leaves vectors unchanged

For every column vector $\mathbf{x} \in \mathbb{R}^n$, $I_n\mathbf{x} = \mathbf{x}$.
Indeed, the columns of the identity matrix from Identity matrix are $\mathbf{e}_1, \ldots, \mathbf{e}_n$. Therefore, writing $\mathbf{x} = (x_1, \ldots, x_n)$,

$$I_n\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \mathbf{x}.$$

Geometric matrix actions in $\mathbb{R}^2$: building intuition

In $\mathbb{R}^2$, a matrix $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 \end{bmatrix}$ sends $\mathbf{e}_1$ to $\mathbf{a}_1$ and $\mathbf{e}_2$ to $\mathbf{a}_2$. Every vector $\mathbf{x} = (x_1, x_2)$ can be written as $x_1\mathbf{e}_1 + x_2\mathbf{e}_2$, since

$$x_1\mathbf{e}_1 + x_2\mathbf{e}_2 = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{x}.$$

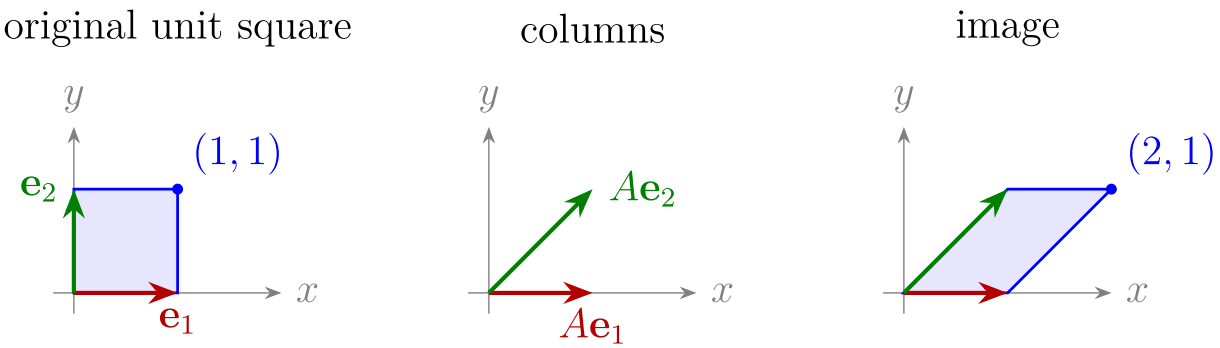
Therefore,

$$\begin{aligned} A\mathbf{x} &= A(x_1\mathbf{e}_1 + x_2\mathbf{e}_2) \\ &= x_1 A\mathbf{e}_1 + x_2 A\mathbf{e}_2 \\ &= x_1\mathbf{a}_1 + x_2\mathbf{a}_2. \end{aligned}$$

The second equality uses linearity, and the last uses the referenced item. Thus a 2×2 matrix is geometrically determined by where it sends the two coordinate directions.

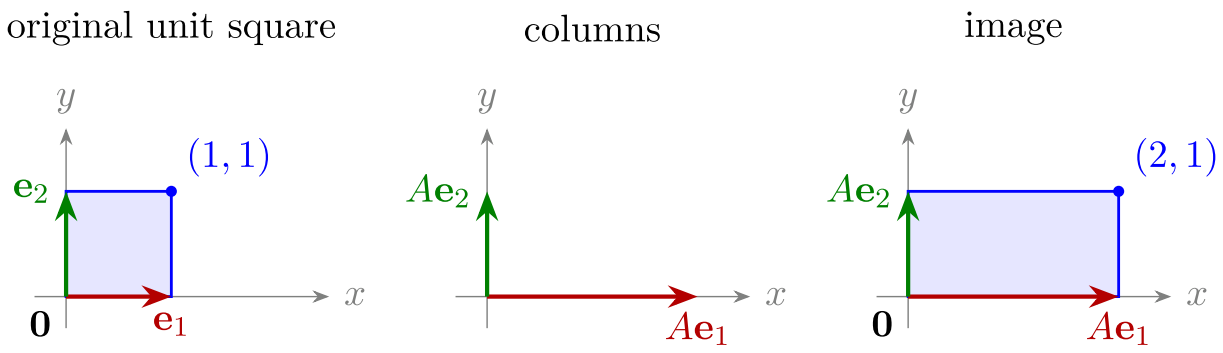
To understand a 2×2 matrix A, apply it to the four corners of the unit square. The image is usually a parallelogram. The two edges leaving the origin are the columns of A. We will refer to this as

the **unit-square visualization**.



$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Figure. A 2×2 matrix sends the unit square to the parallelogram determined by its two columns. This is the geometric version of columns contribute.



$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

Figure. A horizontal stretch by a factor of 2 sends the unit square to a rectangle twice as wide. The first column doubles the horizontal basis vector, while the second column leaves the vertical basis vector unchanged.

- **Matrix:** $\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$

  Ae_1**:** $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$

  Ae_2**:** $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

  Name: horizontal stretch

  Unit-square effect: The right edge moves farther right.

- **Matrix:** $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

  Ae_1**:** $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$

  Ae_2**:** $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

  Name: projection onto the y-axis

  Unit-square effect: Horizontal information is forgotten.

- **Matrix:** $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

  Ae_1**:** $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$

$A\mathbf{e}_2$: $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
Name: reflection across the y-axis
Unit-square effect: The square flips left.

- **Matrix:** $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

$A\mathbf{e}_1$: $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
$A\mathbf{e}_2$: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$
Name: reflection across the line $y = x$
Unit-square effect: The square stays in the same region, but the horizontal and vertical coordinate directions exchange.

- **Matrix:** $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$A\mathbf{e}_1$: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$
$A\mathbf{e}_2$: $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
Name: horizontal shear
Unit-square effect: The bottom edge stays fixed; the top edge shifts right.

- **Matrix:** $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

$A\mathbf{e}_1$: $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
$A\mathbf{e}_2$: $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$
Name: 90° counterclockwise rotation
Unit-square effect: The square rotates around the origin.

- **Matrix:** $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

Ae_1: $\begin{bmatrix}1\\0\end{bmatrix}$

Ae_2: $\begin{bmatrix}0\\0\end{bmatrix}$

Name: projection onto the x-axis

Unit-square effect: Vertical information is forgotten.

Practice recognizing such geometric transformations in Shear Madness.

Activity: Horizontal shear (U1-LO4)

Let $A = \begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}$. Compute Ae_1, Ae_2, and $A\mathbf{x}$, where $\mathbf{x} = (1, 1)$.

We have

$$Ae_1 = \begin{bmatrix}1\\0\end{bmatrix}, \qquad Ae_2 = \begin{bmatrix}1\\1\end{bmatrix}, \qquad A\mathbf{x} = \begin{bmatrix}2\\1\end{bmatrix}.$$

The formula is $A(x, y) = (x + y, y)$. Points with $y = 0$ stay fixed, and points with $y = 1$ shift right by 1.

Note

For the shear matrix $A = \begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}$, the row view gives

$$A\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}(1,1)\cdot(x,y)\\(0,1)\cdot(x,y)\end{bmatrix} = \begin{bmatrix}x+y\\y\end{bmatrix}.$$

The first row measures horizontal plus vertical, while the second row measures vertical only. The column view gives

$$A\begin{bmatrix} x \\ y \end{bmatrix} = x\begin{bmatrix} 1 \\ 0 \end{bmatrix} + y\begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

so the y-coordinate contributes both upward and rightward motion.

Activity: Reflection across the y-axis (U1-LO4)

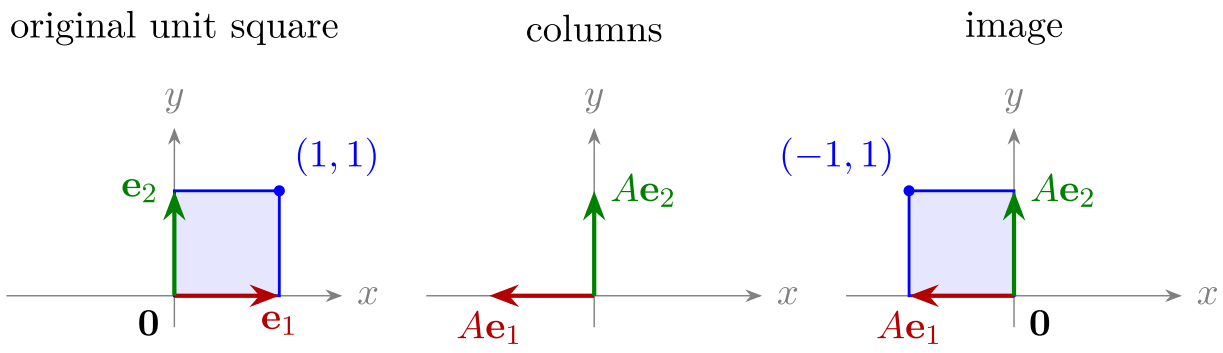
Let $A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$. Compute $A\mathbf{e}_1$, $A\mathbf{e}_2$, $A\mathbf{x}$, and $A\mathbf{y}$, where $\mathbf{x} = (1, 1)$ and $\mathbf{y} = (0, 1)$.

We get

$$A\mathbf{e}_1 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \qquad A\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

$$A\mathbf{x} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \qquad A\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The first coordinate flips sign and the second coordinate is unchanged.



$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Figure. Reflection across the y-axis sends the unit square to its left-hand mirror image. The first column reverses the horizontal direction, while the second column leaves the vertical direction unchanged.

Activity: Reflection across the line y=x (U1-LO4, U1-LO5)

Let

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

1. Compute $A\mathbf{e}_1$ and $A\mathbf{e}_2$.
2. Compute $A \begin{bmatrix} x \\ y \end{bmatrix}$.
3. What happens to a point on the line $y = x$?

We have

$$A\mathbf{e}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \mathbf{e}_2, \qquad A\mathbf{e}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \mathbf{e}_1.$$

More generally, write the input in terms of the

standard basis:

$$A\begin{bmatrix} x \\ y \end{bmatrix} = A(x\mathbf{e}_1 + y\mathbf{e}_2) = xA\mathbf{e}_1 + yA\mathbf{e}_2$$

$$= x\mathbf{e}_2 + y\mathbf{e}_1 = \begin{bmatrix} y \\ x \end{bmatrix}.$$

Thus the two coordinates are exchanged. If $x = y$, exchanging the coordinates does not move the point, so every point on the line $y = x$ stays fixed.

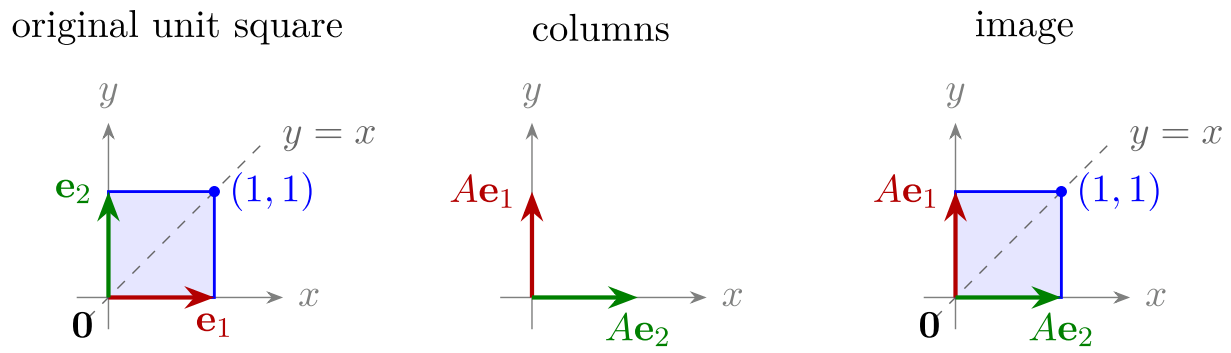


Figure. Three-panel unit-square visualization of reflection across y equals x, with exchanged basis arrows.

Linear and affine maps

A function assigns one output to each input. In this course, many functions have vectors as inputs and vectors as outputs:

$$F : \mathbb{R}^n \to \mathbb{R}^m.$$

The domain of F is its set of allowed inputs, $\mathbb{R}^n$, and its co-domain is the specified set containing its outputs, $\mathbb{R}^m$. A matrix gives one important source of such functions by the rule $\mathbf{x} \mapsto A\mathbf{x}$.

Definition: Matrix map

Given an $m \times n$ matrix A, the matrix map induced by A is the function $T_A : \mathbb{R}^n \to \mathbb{R}^m$ defined by

$$T_A(\mathbf{x}) = A\mathbf{x}.$$

Definition: Linear map

A function $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear if

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}),$$
$$T(c\mathbf{v}) = cT(\mathbf{v})$$

for all vectors $\mathbf{u}, \mathbf{v}$ and all scalars c.

Note: Linear maps preserve linear combinations

If $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear, then for any vectors $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$ and scalars $c_1, \dots, c_k \in \mathbb{R}$,

$$T(c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k) = c_1T(\mathbf{v}_1) + \cdots + c_kT(\mathbf{v}_k).$$

Thus a linear combination of inputs is sent to the linear combination of their outputs with the same coefficients. This follows by repeatedly applying the two linearity rules.

Theorem: Matrix maps are linear

If A is an $m \times n$ matrix, then the function

$$T_A : \mathbb{R}^n \to \mathbb{R}^m, \qquad T_A(\mathbf{x}) = A\mathbf{x}$$

is linear.

Why is this true?

Matrix-vector multiplication distributes over vector addition and scalar multiplication:

$$A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}, \qquad A(c\mathbf{v}) = cA\mathbf{v}.$$

These are exactly the two linearity rules.

Every vector $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ can be written as a linear combination of the standard basis vectors. Indeed, $x_j\mathbf{e}_j$ has entry x_j in position j and zeros elsewhere, so adding these vectors coordinate by coordinate gives

$$x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \mathbf{x}.$$

Since a linear map preserves linear combinations, its values on $\mathbf{e}_1, \dots, \mathbf{e}_n$ determine its value on every input. This observation leads to the following theorem.

Theorem: Every linear map has a matrix

Every linear map $T : \mathbb{R}^n \to \mathbb{R}^m$ is given by multiplication by a unique matrix. The columns of that matrix are $T(\mathbf{e}_1), \dots, T(\mathbf{e}_n)$.

Why is this true?

Using the standard-basis expansion above, let $A = [\, T(\mathbf{e}_1) \ \cdots \ T(\mathbf{e}_n) \,]$. By linearity and the column view of matrix-vector multiplication,

$$
\begin{aligned}
T(\mathbf{x}) &= T(x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n) \\
&= x_1T(\mathbf{e}_1) + \cdots + x_nT(\mathbf{e}_n) \\
&= \begin{bmatrix} T(\mathbf{e}_1) & \cdots & T(\mathbf{e}_n) \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \\
&= A\mathbf{x}.
\end{aligned}
$$

The second equality uses linearity, and the third uses the column view of matrix-vector multiplication. This matrix is unique because its j-th column must be $A\mathbf{e}_j = T(\mathbf{e}_j)$.

Activity: Images of basis vectors determine the matrix (U1-LO5)

Suppose a linear map $T : \mathbb{R}^2 \to \mathbb{R}^2$ satisfies $T(\mathbf{e}_1) = (2, 1)$ and $T(\mathbf{e}_2) = (-1, 3)$. Find the matrix A such that $T(\mathbf{x}) = A\mathbf{x}$.

The columns are the images of the standard basis vectors, so

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}.$$

Definition: Linear function

A linear function is a scalar-valued linear map $f : \mathbb{R}^n \to \mathbb{R}$. Its output is a single real number.

Note: Linear functions are dot products

By Every linear map has a matrix, every linear function is represented by a $1 \times n$ matrix. Writing this row matrix as $\mathbf{c}^T$, where $\mathbf{c} \in \mathbb{R}^n$ is a fixed column vector, gives

$$\begin{aligned} f(\mathbf{x}) &= \begin{bmatrix} c_1 & \cdots & c_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \\ &= c_1 x_1 + \cdots + c_n x_n \\ &= \mathbf{c}^T \mathbf{x} = \mathbf{c} \cdot \mathbf{x}. \end{aligned}$$

Conversely, every function of this form is linear by Matrix maps are linear. Thus linear functions are exactly dot products with fixed vectors. The vector $\mathbf{c}$ is uniquely determined by the values of f on the standard basis vectors. Since $\mathbf{e}_i$ has a 1 in its ith coordinate and 0 in every other coordinate,

$$\begin{aligned} c_i &= c_1(0) + \cdots + c_i(1) + \cdots + c_n(0) \\ &= f(\mathbf{e}_i), \qquad i = 1, \ldots, n. \end{aligned}$$

Example: A linear function in three variables

The function $f : \mathbb{R}^3 \to \mathbb{R}$ given by

$$f(x_1, x_2, x_3) = 2x_1 - x_2 + 3x_3$$

is linear. Indeed, with the fixed column vector $\mathbf{c} = (2, -1, 3)$,

$$f(\mathbf{x}) = \begin{bmatrix} 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \mathbf{c} \cdot \mathbf{x}.$$

Its values on the standard basis vectors recover the entries of $\mathbf{c}$:

$$f(\mathbf{e}_1) = 2, \qquad f(\mathbf{e}_2) = -1, \qquad f(\mathbf{e}_3) = 3.$$

Warning

We use “linear map” and “linear transformation” interchangeably; “transformation” is often used when emphasizing geometry. We reserve “linear function” for scalar-valued linear maps.

Definition: Affine map

Let A be an $m \times n$ real matrix and let $\mathbf{b} \in \mathbb{R}^m$ be a fixed column vector. The function

$$F : \mathbb{R}^n \to \mathbb{R}^m, \qquad F(\mathbf{x}) = A\mathbf{x} + \mathbf{b},$$

where $\mathbf{x} \in \mathbb{R}^n$ is a column vector, is called affine. It is built from a linear map followed by a translation.

An affine map is linear only when $\mathbf{b} = \mathbf{0}$. A quick test is the zero vector: every linear map sends $\mathbf{0}$ to $\mathbf{0}$, but

$$A\mathbf{0} + \mathbf{b} = \mathbf{b}.$$

If $\mathbf{b} \neq \mathbf{0}$, the map is not linear.

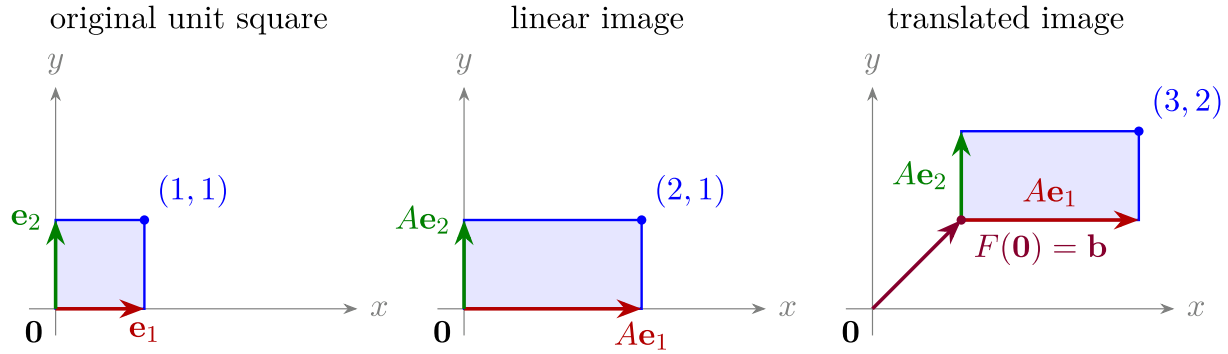


Figure. A unit square is stretched horizontally and then translated one unit right and one unit up.

Example: An affine layer

A layer in a neural network can take the form

$$\mathbf{x} \mapsto W\mathbf{x} + \mathbf{b}.$$

The symbol $\mapsto$ means "maps to": the input vector $\mathbf{x}$ is sent to the output vector $W\mathbf{x} + \mathbf{b}$. The matrix W mixes the input coordinates. The vector $\mathbf{b}$ shifts the result. Since

$$W\mathbf{0} + \mathbf{b} = \mathbf{b},$$

this map is affine. It is linear exactly when $\mathbf{b} = \mathbf{0}$.